

Module 06 - Forecasting and Early Warning Technologies

Hands-on Exercise: Community-Level Flood Early Warning System Assessment

Selected system: Malaysia PublicInfoBanjir / National Flood Forecasting and Warning Centre (PRABN)

Focus: Penang-Kedah community context

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Assignment requirement

The Module 6 Topic 4 exercise requires the student to identify a local Flood Early Warning System, document its components, critique its community-level efficacy, recommend improvements that address identified weaknesses, and save the report as a PDF. This report follows that structure.

1. Selected Flood Early Warning System

The system assessed is Malaysia's official PublicInfoBanjir platform and the National Flood Forecasting and Warning Centre (PRABN) operated by the Department of Irrigation and Drainage Malaysia (JPS/DID). This is an appropriate local system because PublicInfoBanjir provides real-time hydrological monitoring nationwide, including water-level and rainfall station information for both Pulau Pinang and Kedah. PRABN states that its objectives include real-time hydrological monitoring for the public and flood forecasting and warning services for disaster-management agencies.

Scope note: this report evaluates the publicly observable system design and service coverage. It does not claim to measure actual warning receipt, evacuation compliance, or mortality reduction in a specific Penang/Kedah community because no household survey or post-event response dataset was provided in the course materials.

2. System Components

Component	Observed design	Role in EWS
Monitoring / sensors	River water-level and rainfall observations are published through PublicInfoBanjir; the portal also exposes related observation categories such as streamflow, camera and siren information.	Provides the detection layer of the EWS and supports situational awareness.
Data processing and forecasting	PRABN develops and maintains flood forecasting/warning systems including equipment, modelling and ICT infrastructure. It states a target of monsoon flood forecasts/warnings up to two days ahead in river basins where forecast models are complete, subject to	Converts observations and meteorological inputs into forecast/warning information.

	meteorological data.	
Threshold / decision logic	The PublicInfoBanjir water-level legend uses Normal, Alert, Warning and Danger status. Danger indicates considerable flooding and possible evacuation; Warning indicates water nearing flood level and preparation for evacuation; Alert indicates a significant rise above normal.	Transforms hydrological conditions into actionable severity levels.
Dissemination	PublicInfoBanjir publishes current alerts and flood warnings on the web. A registration service supports email/SMS forecast alerts for selected Points of Interest. PublicInfoBanjir also displays siren telemetry in supported areas.	Provides multiple pathways from technical warning to agencies and the public.
User response / last mile	The status definitions explicitly connect levels with preparedness and evacuation actions. Local disaster agencies, communities, schools, businesses and households must then interpret and act on the warning.	Completes the EWS chain; technical detection alone does not reduce risk unless people respond.

3. Community-Level Efficacy: Strengths

- Nationwide real-time visibility. The official portal lists monitoring for Penang and Kedah, so local users can inspect current hydrological conditions rather than relying only on delayed news reports.
- Clear severity language. The four-level water-stage scheme links hydrological status to increasingly urgent protective action, which can reduce ambiguity for operators and communities.
- Multi-layer institutional design. PRABN combines monitoring, modelling, ICT and dissemination rather than treating the forecast model as the entire EWS.
- Multiple dissemination options. Web publication, email/SMS registration for supported POIs, and siren telemetry demonstrate redundancy across digital and physical channels.
- Operational orientation. The course material stresses end-to-end testing, threshold calibration, data quality control, and redundancy; the national system architecture is compatible with these operational principles.

4. Community-Level Efficacy: Weaknesses and Risks

Weakness / risk	Assessment
Uneven forecast-alert coverage	The official registration page states that flood forecast warning registration is currently limited to selected POI locations in the Kelantan, Terengganu and Pahang river basins. In contrast, Penang and Kedah appear in nationwide monitoring. Therefore, monitoring coverage should not be assumed to equal forecast-alert subscription coverage for the local focus area.
Last-mile uncertainty	A portal can publish correct warnings without ensuring that elderly residents, people with limited digital access, migrants, tourists or people asleep at

	night receive and understand them. Community-level efficacy therefore depends on message reach, language, accessibility and trusted local intermediaries.
Static thresholds may not equal impact	A water-level threshold is easy to interpret but does not automatically encode local inundation depth, road closure, school/hospital exposure, tide/backwater effects or drainage congestion. The same river stage can produce different impacts under different antecedent and coastal conditions.
Data-quality and outage risk	The course material notes that real sensor networks can contain gaps, outliers, transmission failures and timestamp errors. During severe weather, communication and power failures may occur exactly when the system is most needed.
Evidence gap for community outcomes	The public interface shows system state and warnings, but it does not by itself demonstrate household receipt rate, behavioural response time, evacuation completion, false-alarm tolerance or trust. Those outcomes need event-based evaluation and community research.

5. Recommended Improvements

Expand forecast-alert POI coverage and geofenced subscriptions: Extend forecast alert registration beyond the currently supported basins as validated local forecast models become available. Allow users to subscribe by home/work/school location and receive geofenced warnings. This directly addresses the gap between nationwide monitoring and uneven forecast-alert coverage.

Adopt multi-channel, multilingual last-mile dissemination: For high-risk areas, combine web/app updates with cell broadcast or SMS, sirens, radio, local authority messaging, mosque/community loudspeakers where appropriate, and institutional relays to schools, hospitals and care homes. Issue concise Malay and English messages, with additional locally relevant languages where needed. This reduces dependence on a single network and improves accessibility.

Move toward impact-based warning: Link forecast water levels and rainfall to expected inundation depth, road disruption, critical-facility exposure and evacuation constraints. A message such as “road X likely impassable within 60 minutes” is more actionable than a stage value alone.

Implement explicit data-quality gating and failover: Automate range checks, spike detection, gap detection, UTC alignment and suppression of alerts when data quality is insufficient. Use upstream proxy gauges, redundant communications and backup power. This addresses false alarms or missed warnings caused by sensor/network failure.

Back-test the complete alert chain: Replay multiple historical flood events through ingestion, model, threshold, classification and dissemination stages. Track POD, FAR, CSI, lead time and P95 latency. Recalibrate thresholds after model updates and after significant events.

Measure community response, not only technical performance: After floods and drills, survey whether warnings were received, understood, trusted and acted upon. Record time-to-action, evacuation bottlenecks, accessibility failures and message comprehension. These indicators reveal whether the EWS actually reduces risk at community level.

6. Proposed Local Operational Workflow

1. Continuous rainfall and river-level monitoring with automated quality control.
2. Forecast/model update using meteorological inputs and current basin state.
3. Alert-level assignment using calibrated thresholds plus impact information.
4. Automated dissemination through at least two independent channels; critical alerts also sent to local authorities and facility operators.
5. Community protective action using predefined SOPs, evacuation routes and shelter information.
6. Post-event review comparing predicted alert timeline with observed flooding, disruptions and response outcomes; thresholds and SOPs are revised if necessary.

7. Conclusion

Malaysia's PublicInfoBanjir/PRABN provides a credible national foundation for flood early warning because it combines real-time monitoring, forecast/warning functions, severity levels and dissemination services. For a Penang-Kedah community context, the strongest publicly verifiable feature is nationwide hydrological monitoring; the main limitation is that forecast-alert subscription coverage should not be assumed to be equally available in every basin. The priority improvements are therefore broader local forecast coverage, redundant and multilingual last-mile dissemination, impact-based warning, data-quality failover, multi-event back-testing and direct measurement of community response. These changes would align the technical system more closely with the people-centred, end-to-end EWS principles emphasised in the DIGHAZ course.

References

1. DIGHAZ. Module 06 | Lecture 04: Forecasting and Early Warning Technologies - System Implementation and Testing. Uploaded course PDF.
2. Department of Irrigation and Drainage Malaysia (JPS/DID). PublicInfoBanjir - National Flood Forecasting and Warning Centre (PRABN): objectives, functions and service charter. <https://publicinfobanjir.water.gov.my/mengenai-kami/prabn/?lang=en>
3. Department of Irrigation and Drainage Malaysia (JPS/DID). PublicInfoBanjir - River Water Level Data and status definitions. <https://publicinfobanjir.water.gov.my/aras-air/data-paras-air/?lang=en>
4. Department of Irrigation and Drainage Malaysia (JPS/DID). PublicInfoBanjir - Flood Alert Registration. <https://publicinfobanjir.water.gov.my/perkhidmatan/daftar-amaran-banjir/?lang=en>
5. Department of Irrigation and Drainage Malaysia (JPS/DID). PublicInfoBanjir - Water Level state coverage including Kedah and Pulau Pinang. <https://publicinfobanjir.water.gov.my/aras-air/>